WORKSHEET · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

Lesson 4: Gravity: What Goes Up...

Module 2 | 45 minutes | Velocity and gravity | Grades 3–5 · ages 9–11

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. velocity ___ A. The lowest place the sprite can stand.
b. gravity ___ B. One step of the game loop (about 30 per second).
c. frame ___ C. A pull that keeps changing the velocity downward.
d. floor ___ D. Speed with a direction (up is +, down is −).

2. Plan your code

Before you build it: Gravity: the Hero starts high up. Use a vy variable: forever change vy by -1 and change y by vy. If y position < -120, set y to -120 and vy to 0, so the Hero lands on the floor. Write or draw the blocks you will use.

3. Follow the velocity

vy is -3. Gravity changes vy by -1 each frame. What is vy two frames later? Circle one.

a. A. -5
b. B. -1
c. C. -3

4. Plan your code

Before you build it: Walk and fall: keep your gravity, and add smooth walking — forever, if right arrow pressed change x by 6; if left arrow pressed change x by -6. Write or draw the blocks you will use.

5. Explain your thinking

Explain in your own words why the sprite falls faster and faster. Use the words velocity and gravity.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

- ☐ I can use a velocity variable.
- ☐ I can code gravity that speeds up a fall.
- ☐ I can make a sprite land on the floor.